

Topology Bootcamp: Week 6

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1 Orientations

We review the interplay between orientation and (co)homology. I'll be adapting this exposition from Hatcher chapter 3.

As you may have seen in your algebraic topology class, the classic process for defining orientations on manifolds is as follows:

- (1) First, we define orientations for $\mathbb{R}^n$.
- (2) Then, since manifolds are locally $\mathbb{R}^n$, we can define local orientations at any point in our manifold.
- (3) Imposing some coherence conditions, we are able to extend this notion globally.

Any notion of “orientation” of $\mathbb{R}^n$ should have the property that it is invariant under rotation but reverses under reflection. In $\mathbb{R}^2$, the notion of “clockwise” and “counterclockwise” has this property, as do “left-handed” and “right-handed” in $\mathbb{R}^3$. Another one you may have seen is to choose some ordered basis $\mathcal{B} = \{e_i\}_{i=1}^n$ for $\mathbb{R}^n$ (or really any finite-dimensional vector space). For any other choice $\mathcal{B}' = \{e'_i\}_{i=1}^n$, there is a unique linear transformation φ sending $e_i \mapsto e'_i$. If φ has positive determinant, we say that $\mathcal{B}$ and $\mathcal{B}'$ have the same orientation; otherwise they have opposite orientations. In particular, any rotation is orientation-preserving (has positive determinant) and any reflection is orientation-reversing (has negative determinant), as desired.

If our goal were specifically define orientation for *smooth* manifolds M , this would be a sufficient definition of orientation for $\mathbb{R}^n$ (Step (1)). Since we have a canonical vector space at each point $x \in M$, the *tangent space* $T_x M$, a *local orientation* might be a choice of orientation of $T_x M$ (Step (2)). Moreover, it is straightforward to extend this globally. We want some way of “varying” this orientation smoothly, and the way we *always* do that for infinitesimally-defined data is to interpret it as a section of TM . So, to complete Step (3), define an *orientation* of M to be a choice of n linearly-independent sections of TM . This induces a local orientation at each point x in M , and if we have two charts U_1, U_2 and points $x_1, x_2 \in U_1, U_2$, the transition map $\varphi_{1,2}$ induces a linear map $D_{x_1} \varphi_{1,2} : T_{x_1} M \rightarrow T_{x_2} M$, which you can check is orientation preserving.

However, for general manifolds, we have none of this smooth machinery to work with; all we have is the locally Euclidean property. As such, we need a way to define orientations at a point in $\mathbb{R}^n$ such that this notion can be extended to a ball, so that we might complete Step (3). This is where homology comes in.

Definition 1.1. A *local orientation* of M at a point x is a choice of generator μ_x of $H_n(M, M \setminus \{x\}) \cong \mathbb{Z}$.

Indeed, to see if this definition satisfies our characteristic property of orientations, note that

$$H_n(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong H_{n-1}(\mathbb{R}^n \setminus \{x\}) \cong H_{n-1}(S^{n-1})$$

where S^{n-1} is the sphere centered at x . A rotation ρ induces a map of degree 1 while a reflection R is of degree -1, and so for a generator α , $\rho_*(\alpha) = \alpha$ while $R_*(\alpha) = -\alpha$. This extends to all of $\mathbb{R}$, since for any

other point y and some large ball B containing both x and y , there is a canonical isomorphism

$$H_n(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong H_n(\mathbb{R}^n, \mathbb{R}^n \setminus B) \cong H_n(\mathbb{R}^n, \mathbb{R}^n \setminus \{y\}).$$

Let the local orientation at y be the image of α .

Now, let M be any n -manifold. By excision,

$$H_n(M, M \setminus \{x\}) \cong H_n(B, B \setminus \{x\}) \cong H_n(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong \mathbb{Z},$$

where B is some ball containing x under the image of a chart. So, we can define

Definition 1.2. A *local orientation* of M at a point x is a choice of generator μ_x of $H_n(M, M \setminus \{x\})$.

Topological manifolds also have transition homeomorphisms, and we say that transitions are *orientation preserving* if the induced isomorphisms on local homology fixes generators, and *orientation reversing* otherwise. A choice of charts where all transition functions are orientation preserving is called a *orientation atlas*, and

Definition 1.3. An *orientation* on M is a choice of maximal orientation atlas. If an orientation exists, M is *orientable*.

Every manifold M has an orientable two-sheeted covering space $\widetilde{M}$. We can define the *orientation double cover*

$$\widetilde{M} = \{\mu_x : x \in M \text{ and } \mu_x \text{ is a local orientation of } M \text{ at } x\}.$$

The map $\mu_x \mapsto x$ gives a 2:1 surjection $\widetilde{M} \rightarrow M$, and there is a topology we can put on $\widetilde{M}$ that makes this map a covering space map (see Hatcher if you care). In particular,

Proposition 1.1. If M is connected, then M is orientable if and only if $\widetilde{M}$ has two components. Moreover, M is orientable if it is simply connected, or more generally if $\pi_1(M)$ has no subgroup of index 2.

There is actually a larger covering space $M_{\mathbb{Z}}$ which $\widetilde{M}$ embeds into. As a set, $M_{\mathbb{Z}}$ consists of all $\alpha_x \in H_n(M, M \setminus \{x\})$ for all $x \in M$. Then we get a covering space map $M_{\mathbb{Z}} \rightarrow M$ which is infinite-sheeted.

Definition 1.4. A continuous map $M \rightarrow M_{\mathbb{Z}}$ of the form $x \mapsto \alpha_x \in H_n(M, M \setminus \{x\})$ is a *section* of $M_{\mathbb{Z}}$.

An orientation of M is the same thing as a section of $M_{\mathbb{Z}}$ where $x \mapsto \mu_x$ such that μ_x is a generator for all x .

More generally, we can consider the homology groups with coefficient groups over a commutative unital ring R . Then we can construct a covering space $M_R \rightarrow M$, and

Definition 1.5. An *R -orientation* of M is a section of M_R sending x to a “generator” of

$$H_n(M, M \setminus \{x\}, R) \cong R$$

for all x . Here, a generator of R is an element u such that $Ru = R$, and since R has an identity, u is a unit.

Recall,

$$H_n(M, M \setminus \{x\}; R) \cong H_n(M, M \setminus \{x\}) \otimes R.$$

From this, we get a very concrete description of M_R . Each $r \in H_n(M, M \setminus \{x\}; R) \cong R$ determines a subcovering space M_r of M_R consisting of the points $\pm \mu_x \oplus r \in H_n(M, M \setminus \{x\}; R)$, where μ_x is a generator of $H_n(M, M \setminus \{x\}) \cong \mathbb{Z}$. If r has order 2, so $r = -r$, then M_r is just a copy of M , and otherwise M_r is isomorphic to $\widetilde{M}$. The space M_R is the union of all M_r , which are disjoint except for the equality $M_r = M_{-r}$.

It follows that

Proposition 1.2. An orientable manifold is R -orientable for all R , while a non-orientable manifold is R -orientable if and only if R contains a unit of order 2, which is equivalent to $2 = 0$ in R . Therefore, every manifold is $\mathbb{Z}/2$ -orientable. In practice, this means that $R = \mathbb{Z}$ and $R = \mathbb{Z}/2$ are the most important cases to consider.

Theorem 1.1. Let M be a closed connected n -manifold. Then

- (1) If M is R -orientable, the map $H_n(M; R) \rightarrow H_n(M, M \setminus \{x\}; R) \cong R$ is an isomorphism for all x .
- (2) If M is not R -orientable, this map is injective with image $\{r \in R : 2r = 0\}$ for all $x \in M$.
- (3) $H_i(M; R) = 0$ for all $i > n$.

In particular, $H_n(M; \mathbb{Z})$ is either $\mathbb{Z}$ or 0, depending on whether M is orientable or not. In either case, $H_n(M; \mathbb{Z}/2) \cong \mathbb{Z}/2$.

Definition 1.6. An element of $H_n(M; R)$ whose image in $H_n(M, M \setminus \{x\}; R)$ is a generator for all x is called a *fundamental class* for M with coefficients in R .

By the above theorem, a fundamental class exists if M is closed and R -orientable. Indeed, the converse holds: given a fundamental class $\mu \in H_n(M; R)$, let μ_x be its image in $H_n(M, M \setminus \{x\}; R)$. The function $x \mapsto \mu_x$ is an R -orientation since $H_n(M; R) \rightarrow H_n(M, M \setminus \{x\}; R)$ factors through $H_n(M, M \setminus B; R)$ for any open ball $B \ni x$. Furthermore, M must be compact since μ_x can only be nonzero for x in the image of a cycle representing μ , and this image is compact.

2 Poincaré Duality

Recall from Aislinn and Peter's talk,

Theorem 2.1. If M is a closed R -orientable n -manifold with fundamental class $[M] \in H_n(M; R)$, then the map $D : H^k(M; R) \rightarrow H_{n-k}(M; R)$ given by $D(\alpha) = [M] \frown \alpha$ is an isomorphism for all k .

3 Poincaré-Lefschetz Duality

It turns out that Poincaré duality is a special case of a more general duality for manifolds with boundary.

Definition 3.1. A compact manifold M with boundary is defined to be *R -orientable* if $M \setminus \partial M$ is R -orientable as a manifold without boundary.

When M is R -orientable, there is a relative fundamental class $[M, \partial M] \in H_n(M, \partial M; R)$ which restricts to a given orientation at each point of $M \setminus \partial M$.

Theorem. (Hatcher 3.43). If M is an R -orientable n -manifold with boundary and $\partial M = A \cup B$, where A, B are $(n-1)$ -manifolds with $\partial A = \partial B = A \cap B$, then

$$D_M : H^k(M, A; R) \rightarrow H_{n-k}(M, B; R)$$

given by $\alpha \mapsto [M] \smile \alpha$ is an isomorphism for all k .

Proof. The cap product map D_M is well defined by a standard alg top prelim argument (there exists open neighborhoods which deformation retract via the collar neighborhood theorem). If $B = \emptyset$, then in fact this result follows from ordinary Poincaré duality for $M \setminus \partial M$ (in practice, this is all we'll really need). $\square$

4 Intersection Form for Manifolds with Boundary

Let X be a compact, orientable, topological 4-manifold. Let $[X, \partial X]$ be a fundamental class of X in $H_4(X, \partial X; \mathbb{Z})$.

Definition 4.1. The symmetric bilinear form

$$Q_X : H^2(X, \partial X; \mathbb{Z}) \times H^2(X, \partial X; \mathbb{Z}) \rightarrow \mathbb{Z}$$

defined by $Q_X(a, b) = \langle a \smile b, [X, \partial X] \rangle = a \cdot b$ is called the *intersection form* of X . Since by Poincaré-Lefschetz duality $H_2(X) \cong H^2(X, \partial X)$, we have that Q_X is defined on $H_2(X) \times H_2(X)$ as well.

Note that this only requires the topological structure of X to define. Moreover, it's clear (if not, an exercise) that if a or b are torsion, then $Q_X(a, b) = 0$, and so Q_X descends to a pairing

$$Q_X : H_2(X)/\text{torsion} \times H_2(X)/\text{torsion} \rightarrow \mathbb{Z}.$$

So, if we choose a basis of $H_2(X)/\text{torsion}$, we can represent Q_X as an integral matrix M , where

$$Q_X(a, b) = \alpha^T M \beta$$

where α, β are Poincaré dual to a, b .

Some remarks:

- (1) Q_X can be extended to cohomology with any coefficient ring R (and if X is not orientable, we only need that X is R -orientable). That is for *any* manifold (possibly non-orientable), we can always speak about the $\mathbb{Z}/2$ intersection form.
- (2) We can actually define another pairing $Q'_X : H^2(X) \times H^2(X, \partial X) \rightarrow \mathbb{Z}$ given by $Q'_X(a, b) = \langle a \smile b, [X] \rangle$.

A super slick proof to be aware of:

Proposition 4.1. Let X be a closed, oriented, smooth 4-manifold. Then every element of $H_2(X)$ can be represented by an embedded surface.

Proof. Elements of $H^2(X)$ are in bijective correspondence with $U(1)$ -bundles over X . For $\alpha \in H^2(X)$, take its Poincaré dual $a \in H^2(X)$ and denote the corresponding $U(1)$ -bundle by $L_\alpha \rightarrow X$. The zero set of a generic section of the bundle $L_\alpha \rightarrow X$ will be a smooth surface representing α . $\square$

Proposition 4.2. For $a, b \in H^2(X)$ and $\alpha, \beta \in H_2(X)$ their Poincaré dual, $Q_X(a, b)$ is the signed count of points in $\Sigma_\alpha \cap \Sigma_\beta$.

Proof. Let $\omega = a \otimes \mathbb{R}$ and $\eta = b \otimes \mathbb{R}$, so $\omega, \eta \in H^2(X; \mathbb{R})$. Under the de Rham isomorphism, we can think of ω and η as exterior 2-forms on X . Let $\Sigma_\alpha, \Sigma_\beta$ be smooth representatives of α, β , and suppose that ω, η are supported in a small tubular of $\Sigma_\alpha, \Sigma_\beta$ (see e.g. the construction of the Thom form). Construct ω, η so that in a coordinate chart (x, y, u, v) around each intersection point $p \in \Sigma_\alpha \cap \Sigma_\beta$ where Σ_α is given by $\{x = 0, y = 0\}$ and Σ_β by $\{u = 0, v = 0\}$, we have that

$$\omega = f(x, y) dx \wedge dy, \quad \text{and} \quad \eta = g(u, v) du \wedge dv$$

where f is a bump function localized around $0 \in \mathbb{R}^2$ with integral 1. It is straightforward to see that $Q_X(a, b) = \int_X \omega \wedge \eta$, and the assertion follows. $\square$

Theorem 4.1. If X is simply connected (e.g. admits a handle diagram with no 1-handles), Q_X is unimodular if and only if ∂M is a disjoint union of integer homology spheres.

Proof. The forward implication is straightforward from Poincaré-Lefschetz duality, as $H^2(M) = H^2(M, \partial M)$, and $Q'_X : H^2(M) \times H^2(M, \partial M) \rightarrow \mathbb{Z}$ is non-degenerate.

Assume without loss of generality that ∂M is connected. The main observation for the reverse implication is that $H_3(M, \partial M) = H^1(M) = 0$. Observe, the map $\iota_* : H_2(\partial M) \rightarrow H_2(M)$ induced by inclusion must have empty image, as otherwise Q_X would be degenerate (pairing with anything in the image gives 0). Thus, we have an exact sequence¹

$$0 \rightarrow H_2(\partial M) \xrightarrow{\iota_*} H_2(M) \rightarrow H_2(M) \rightarrow H_2(M, \partial M) \rightarrow H_1(\partial M) \rightarrow 0.$$

Then the image of ι_* is empty if and only if $H_2(\partial M) = 0$. Poincaré duality then implies that ∂M is an integer homology sphere. $\square$

Theorem 4.2. Let X be a 4-manifold with boundary obtained by integral surgery on a framed link L in S^3 . Then Q_X is isomorphic to the linking matrix of L . Moreover, it's possible to choose an oriented basis of $H_2(M)$ such that Q_X is *equal* to the linking matrix of L .

Corollary 4.2.1. Let M be a 3-manifold obtained by integral surgery on a framed link L . Then M is an integer homology sphere if and only if the determinant of the linking matrix is ± 1 .

Corollary 4.2.2. every integral symmetric bilinear form Q is the intersection form of a smooth simply-connected 4-manifold M with boundary. If Q is unimodular, then ∂X is a homology sphere.

5 Exercises

- (1) Show that removing a point from a manifold of dimension greater than 1 does not affect orientability.
- (2) Show that every covering space of an orientable manifold is orientable.
- (3) Show that if M_1 and M_2 are closed n -manifolds, then there are isomorphisms

$$H_k(M_1 \# M_2) \cong H_k(M_1) \oplus H_k(M_2)$$

for $0 < k < n$, with one exception: if both M_1 and M_2 are nonorientable, then $H_{n-1}(M_1 \# M_2)$ is obtained from $H_{n-1}(M_1) \oplus H_{n-1}(M_2)$ by replacing one of the two $\mathbb{Z}/2$ summands by a $\mathbb{Z}$ summand (hint: Euler characteristic). Additionally, show that $\chi(M_1 \# M_2) = \chi(M_1) + \chi(M_2) - \chi(S^n)$.

- (4) The if M and N are closed orientable n -manifolds with fundamental classes $[M], [N]$, then the *degree* of a map $f : M \rightarrow N$ is defined to be the integer such that $f_*([M]) = d[N]$. Show that there always exists a degree 1 map $M \rightarrow S^n$.
- (5) Suppose that $Q : \mathbb{Z}^n \times \mathbb{Z}^n \rightarrow \mathbb{Z}$ is some integral bilinear form. Show that Q is even if at least one matrix representation of Q has even diagonal.
- (6) Let $\langle n \rangle$ be a bilinear form over $\mathbb{Z}$ represented by the matrix $[n]$. Prove that $\langle n \rangle$ and $\langle m \rangle$ are equivalent if and only if $m = n$.

¹Note that it's important that we have the collar neighborhood theorem here.